

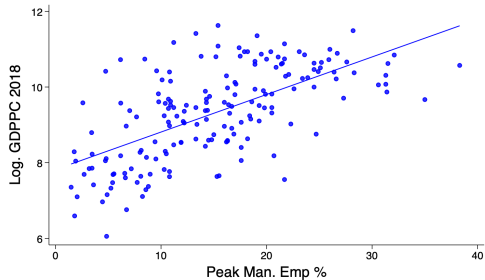
From Agriculture to Services: Development without Industrialization

Tancredi Rapone

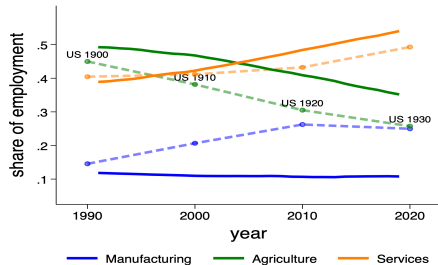
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Barcelona School of Economics (BSE)
Universitat Autònoma de Barcelona (UAB)

Manufacturing Matters — But Many Countries Skip It



(a) Manufacturing peaks and income gaps



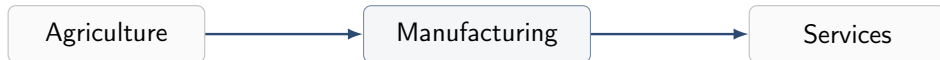
(b) Different paths into services.

Data: ILOEST, Gapminder.

This Paper: Structural Change and Development

The traditional view of development

Kuznets (1966); Chenery and Syrquin (1975); Herrendorf, Rogerson and Valentinyi (2014)



A new development path

Rodrik (2016)



Does industrialization change
the service economy that follows?

Mechanism: Manufacturing Shapes the Service Economy

Large-scale manufacturing

Repeated coordination problems

hire
→

High-skill producer services

Managers, engineers and accountants develop **reusable methods**.

diffuse
→

Capabilities for services

Service firms can adapt these managerial and technical methods.

Historical examples of reusable methods

DuPont

Return-on-investment accounting

General Motors

Multidivisional organization

Hypothesis: Manufacturing builds managerial and technical capabilities that remain useful as the economy moves into services.

Chandler (1977, 1990); McKenna (2006); Mol and Birkinshaw (2014).

Historical Example: Managerial Accounting

1914
DuPont

Manufacturing origin



ROI



1922–24
McKinsey

Codification



1935
Marshall Field's

Transfer to services



McKinsey becomes
CEO

Document

Cross-country stylized facts

Larger manufacturing peaks are associated with more high-skill producer services (HSPS) and faster growth after manufacturing declines.

Identify

U.S. counties: shift-share evidence

Exposure to national growth in high-throughput manufacturing predicts larger industrial peaks and greater HSPS specialization in 2000.

Quantify

Growth model: firm scale and knowledge diffusion

Scale raises the return to formalization. The model reproduces **98%** of the cross-country HSPS gradient and **40%** of the growth gradient.

Policy: Scale reforms strengthen the lasting effects of support for skills and organizational investment.

Literature

- **Structural transformation and service-led development**

Herrendorf et al. (2014); Rodrik (2016); Buera and Kaboski (2012); Buera et al. (2022); Fan et al. (2023); Peters et al. (2026).

⇒ I study how manufacturing history shapes service composition and subsequent growth.

- **Firm scale, skills and organizational capital**

Chandler (1977, 1990); Prescott and Visscher (1980); Atkeson and Kehoe (2005); Engbom et al. (2025).

⇒ I link firm scale to skill demand and the creation of reusable organizational procedures.

- **Industrial policy: big push, increasing returns and persistence**

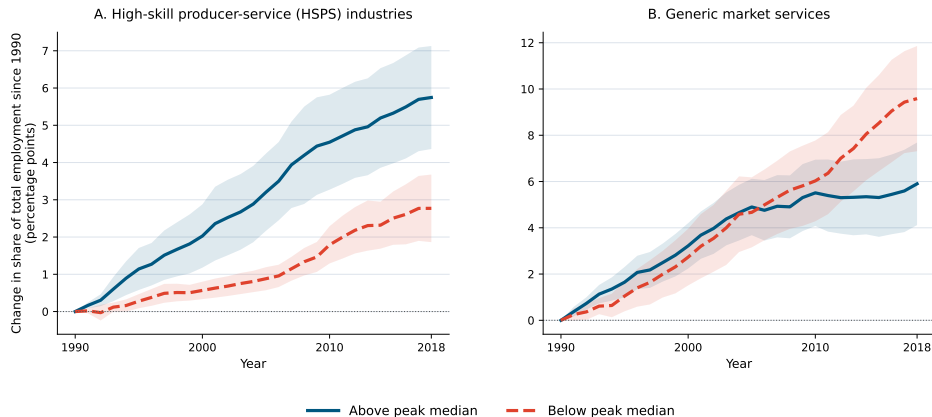
Rosenstein-Rodan (1943); Hirschman (1958); Murphy et al. (1989); Juhász (2018); Incoronato and Lattanzio (2025); Choi and Levchenko (2025).

⇒ I show how industrial policy can have lasting effects **beyond manufacturing**.

1. Cross-country stylized facts
2. U.S. county evidence
3. Model: organization and growth
4. Quantitative analysis
5. Policy implications

Cross-Country Facts

Stylized Fact 1: Manufacturing and Service Composition

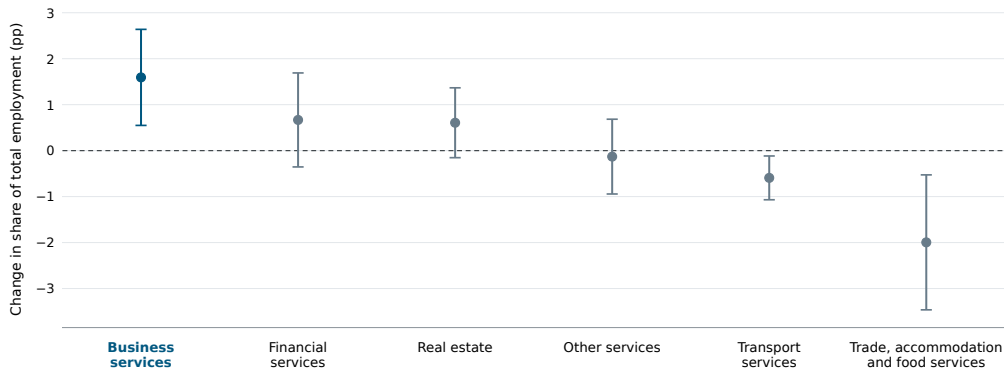


51 economies; median manufacturing-peak split. Employment shares rebased to 1990; unweighted means ± 1.96 SE. HSPS: business services, finance and real estate. Source: ETD and GGDC.

[► Regression table](#)

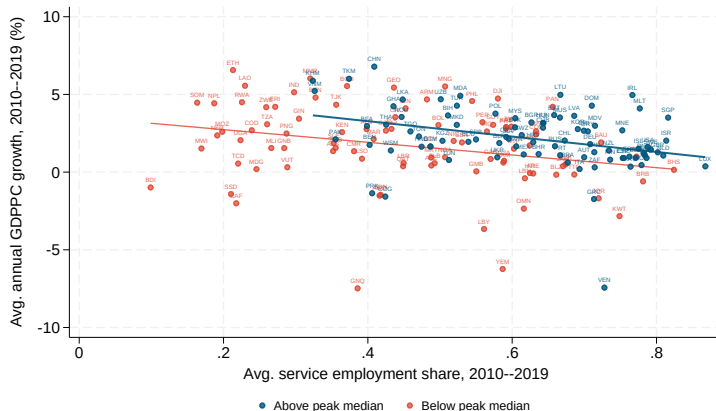
Stylized Fact 1: Service-Industry Decomposition

Change in employment share, 1990–2018, per 10 pp larger manufacturing peak.



51 economies; separate OLS regressions, no additional controls. Shares of total employment; 95% CIs using HC3 standard errors. Source: ETD and GGDC.

Stylized Fact 2: Manufacturing and Service-Era Growth



2010–2019 averages; median manufacturing-peak split. Source: GGDC, ILOSTAT and World Bank/Gapminder.

► Regression table

U.S. County Evidence

Following Industrialization across U.S. Counties

Fixed 1850 county boundaries connect historical Census data to modern employment records.

Object	Measurement
Industrialization	Peak manufacturing wage earners / classified employment, over 1880, 1900, 1920, 1930 and 1940.
Establishment scale	Log manufacturing workers per establishment in the county's peak year.
Inside manufacturing	HSPS occupations : accounting, coordination, personnel, engineering and technical work.
Specialized services	HSPS industries : information; professional, scientific and technical services; management of companies (NAICS 51, 54, 55).

Event-study denominators: classified employment. Cross-sectional denominator: private-service employment. Sources: Census of Manufactures, Population Census, CBP/EFSY and QCEW.

► Definitions and sources

Tracing Capabilities around Manufacturing Peaks

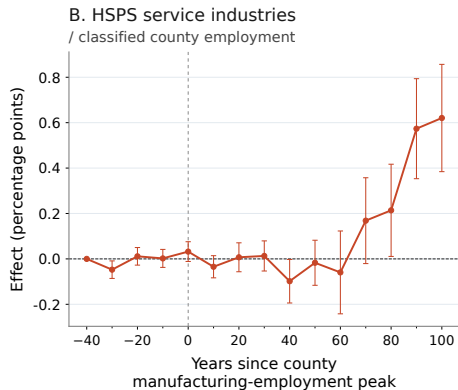
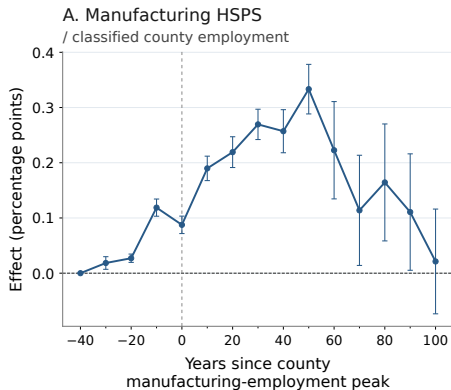
Y_{ct} : manufacturing HSPS or HSPS-service employment, both in % of classified employment.

$$Y_{ct} = \alpha_c + \gamma_{s(c),t} + \lambda_{t-P_c} + \sum_{k \neq -40} \beta_k \left[1\{t - P_c = k\} \frac{M_c^{\text{peak}}}{0.10} \right] + X_c' \delta \text{Trend}_t + \varepsilon_{ct}.$$

- **1,116 counties** peaking in 1900, 1920 or 1940, with a common forty-year pre-period.
- County, state-by-year and event-time fixed effects; baseline covariates interacted with linear calendar-time trends.

P_c : manufacturing-peak year. Reference: $t - P_c = -40$. Conley covariance, 100 km.

Manufacturing Capabilities Precede Specialized HSPS Services



Per 10 pp larger manufacturing peak; reference: -40. 1,116 counties; 95% CIs (Conley, 100 km). Source: Census and harmonized CBP.

► Falsification: all services

► By manufacturing type

Scale, Manufacturing HSPS, and Later Service Composition

Dependent variable: HSPS share of private-service employment in 2000.

	Baseline	+ Scale	+ Mfg. HSPS	+ Both	+ Other channels
Manufacturing peak	0.048*** (0.010)	0.024** (0.012)	0.030*** (0.011)	0.015 (0.012)	0.017* (0.010)
Log workers per manufacturing establishment		0.0060*** (0.0022)		0.0041* (0.0023)	
Manufacturing HSPS / classified employment			2.112*** (0.529)	1.927*** (0.535)	
Attenuation of peak coefficient (%)	0.0	50.3	37.4	68.3	63.9

Shares in fractions. 1,487 counties; state/peak-cohort FE and 1850 controls. Other channels: wages, urbanization and population. Conley SE (100 km). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Census and QCEW.

Which Manufacturing Industries Require Production at Scale?

High-throughput manufacturing

Continuous processing, large batches and complex assembly make scale and coordination integral to production.

Iron and steel; chemicals; machinery; processed foods.

Traditional low-throughput manufacturing

Production can operate efficiently in smaller establishments.

Textiles and apparel; wood and furniture; printing; leather goods.

Prediction: The service-sector legacy should be stronger when manufacturing expansion creates intensive organizational and technical demands.

Classification follows Chandler (1990).

National Industry Growth and Local Industrialization

Combine each county's **1880 industry mix** with subsequent **national industry growth**:

$$Z_c^g = \sum_{j \in g} \underbrace{\omega_{cj,1880}^g}_{\text{1880 industry share}} \underbrace{G_{j,-c,1880-1900}}_{\text{leave-county-out growth}}, \quad g \in \{A, H, L\}.$$

- **Three instruments:** all manufacturing (A), high throughput (H), and traditional low throughput (L).
- **Treatment and outcome:** instrument the manufacturing peak to explain HSPS specialization within services in 2000.
- **Conditioning:** state and peak-cohort fixed effects, 1850 characteristics, and complete 1850–1870 development histories.

Identifying assumption: Conditional on these histories, national industry growth affects later services through the local industrialization episode it induces.

1880 employment shares normalized within each group; standardized shifters. Histories: manufacturing, manufacturing HSPS and HSPS services, 1850–1870.

National Industry Growth and Persistent HSPS Services

Dependent variable: HSPS share of private-service employment in 2000.

	OLS	All industries IV	High-throughput IV	Low-throughput IV
Manufacturing peak	0.029** (0.012)	0.133 (0.105)	0.283** (0.118)	−1.095 (8.010)
First stage, per 1 SD of shifter		0.0150*** (0.0039)	0.0193*** (0.0054)	0.0006 (0.0042)
First-stage Conley F		14.67	12.87	0.02
Kleibergen–Paap F		20.53	21.94	0.03
Anderson–Rubin p -value		0.203	0.009	0.596
Counties	1,487	1,487	1,487	1,487

Shares in fractions. State/peak-cohort FE, 1850 controls and 1850–1870 histories. Conley SE (100 km); AR: Anderson–Rubin. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Census and QCEW.

► Rotemberg weights

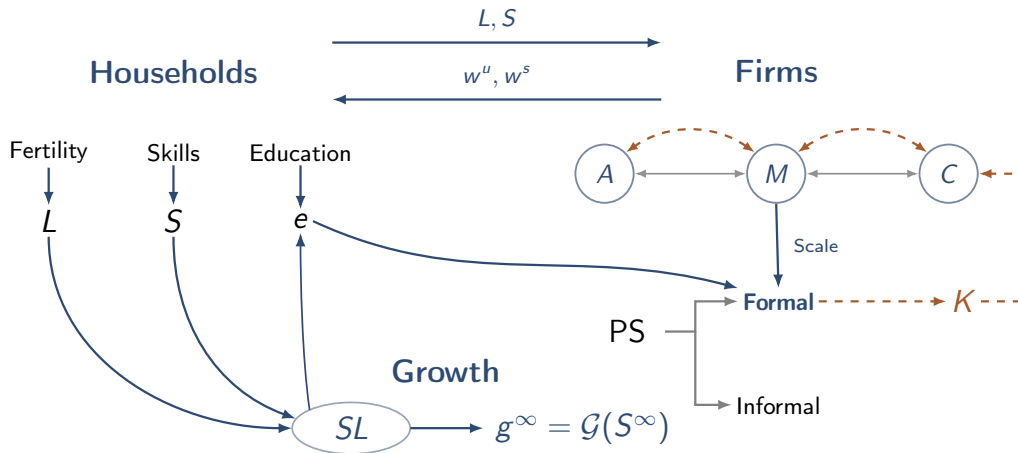
► Measurement

► Shock inference

► Other checks

Model

Model Overview



Three final-good sectors: agriculture, manufacturing, consumer services.

Firm ability $a_f \sim G$; inherited organizational frontier s_{fjt} .

$$v_{fjt} = A_{jt} a_f \ell_{fjt}^\eta, \quad 0 < \eta < 1, \quad n_{fjt} = \nu_{jt} \ell_{fjt}.$$

ℓ : CES composite of skilled and unskilled labor; n : employment headcount.

Gross output

$$y_{fjt} = Z_{jt} v_{fjt}^{\theta_{jt}} (m_{fjt}^a)^{\gamma_{jt}^a} (m_{fjt}^m)^{\gamma_{jt}^m} (P_{fjt}^P)^{\gamma_{jt}^P}, \quad \theta_{jt} + \gamma_{jt}^a + \gamma_{jt}^m + \gamma_{jt}^P = 1.$$

m^a, m^m : materials; P^P : producer services used in production.

Organizational Tasks: Informal or Formal

Task complexity $z \geq 0$ has CDF $\mathcal{F}(z) = 1 - e^{-z}$.

$$\underbrace{q(s) = 1 - e^{-s}}_{\text{formalized task share}}, \quad \underbrace{b(n) = \frac{n}{\bar{N}}}_{\text{problems per task}}.$$

Informal problem solving

$$L^o(n, s) = \frac{n}{\bar{N}} r [1 - q(s)]$$

Low-skill services; repeated resolution,
 $r > 1$.

Formal procedures

$$H^o(n, s) = \frac{n}{\bar{N}} q(s)$$

High-skill services; reusable procedures.

$$C_t^o(n, s) = \frac{n}{\bar{N}} [q(s)p_t^H + r[1 - q(s)]p_t^L].$$

Formalization also changes the high-skill weight in productive producer services P^P .

Firm Scale and the Return to Formalization

$$\pi_{jt}^*(a, q) = \max_{\ell \geq 0} \{ p_{jt}^v(q) A_{jt} a \ell^\eta - [w_{jt}^v + c_{jt}^o(q)] \ell \}.$$

$p^v(q)$: value-added price net of intermediate costs; $c^o(q)$: coordination cost per unit of labor.

Proposition 1: Scale–formalization complementarity

$$\frac{\partial \pi_{jt}^*(a, q)}{\partial q} = \underbrace{n_{jt}^*(a, q)}_{\text{firm employment}} \underbrace{\Psi_{jt}(q)}_{\text{net gain per worker}}.$$
$$\Psi_{jt}(q) = \underbrace{\frac{w_{jt}^v + c_{jt}^o(q)}{\eta \nu_{jt}} \frac{\partial \log p_{jt}^v(q)}{\partial q}}_{\text{productive-service cost effect}} + \underbrace{\frac{rp_t^L - p_t^H}{\bar{N}}}_{\text{coordination saving}}.$$

Conditional on equilibrium prices and $\Psi_{jt}(q) > 0$. If also $rp_t^L > p_t^H$, greater formalization raises desired employment.

► Derivation

Creating and Adapting Organizational Procedures

K_t : share of tasks with a publicly available procedure.

Create a new procedure

$$C_t^C(z) = \frac{p_t^H}{\bar{e}_t^o} e^{z-1}.$$

$\bar{e}_t^o$: education in organizational labor; ξ_j : adaptability; $0 < \rho < 1$, $\xi_m = 1$.

Adapt a public procedure

$$C_{jt}^A(z) = \frac{\rho}{\xi_j} C_t^C(z), \quad \mathcal{F}(z) \leq K_t.$$

One-time investment; future operating returns

$$W_{jt}(a, s) = \pi_{jt}^*(a, s) + \max_{u \geq s} \{ -C_{jt}^O(s, u; K_t) + \beta(1 - \delta_f) W_{j,t+1}(a, u) \}.$$

Current production uses s ; investment installs next period's frontier u .

► Investment and diffusion

Public Knowledge and Equilibrium Firm Scale

Proprietary procedures diffuse

$$K_{t+1} = \min\{1, K_t + \delta_K \mathcal{E}_t\}.$$

$\mathcal{E}_t$: employment-weighted exposure to newly created procedures.

Entry determines average scale

$$V_{jt}^E = \int W_{jt}(a, 0) dG(a), \quad F_j l_{jt}^{\phi_E} = V_{jt}^E \quad \text{if } l_{jt} > 0.$$

$$M_{jt} = (1 - \delta_f) M_{j,t-1} + l_{jt}, \quad \bar{n}_{jt} = \frac{N_{jt}}{M_{jt}}.$$

l : entrants; M : firm mass; F_j : setup-cost shifter. Entrants draw ability after paying and start at $s = 0$.

Public knowledge does not depreciate. Creating firms do not internalize subsequent users' adaptation gains.

Households: Fertility, Education and Consumption

Pooled income and common consumption; employment origin i affects childcare time τ_i .

$$U_{i,t} = (1 - \gamma) \log C_t + \gamma \log(n_{i,t} h_{i,t}), \quad h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t}.$$

Time and resources

$$T_{i,t} = \tau_i + \tau_e(S_t) e_{i,t}, \quad \tau_e(S) = 1 - S(1 - \tau_e^s), \quad 0 < \tau_e^s < 1.$$

$$\sum_j p_{j,t}^c C_{j,t} = (1 - n_{i,t} T_{i,t}) \bar{w}_t + d_t.$$

Nonhomothetic consumption

$$\sum_{j \in \{a, m, c\}} \omega_j^{1/\sigma} \left(\frac{C_{j,t}}{C_t^{\varepsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad 0 < \sigma < 1.$$

$\bar{w}_t$: pooled potential earnings; d_t : dividends. Employment origin $i \in \{a, m, s\}$, with s pooling all services.

Galor and Weil (2000); Lagerlöf (2006); Comin et al. (2021).

Skill Supply, Population and Productivity Growth

Entering cohorts choose skill

$$p_t^S = \mathcal{S}(\Pi_{t+1}), \quad \mathcal{S}' > 0, \quad S_{t+1} = \left(1 - \frac{1}{\Delta}\right) S_t + \frac{1}{\Delta} p_t^S.$$

Π_{t+1} : anticipated skilled-wage premium; Δ : generation length.

Population and sectoral productivity

$$L_{t+1} = L_t \bar{n}_t^{1/\Delta}, \quad g_{j,t}^A = [1 + \lambda_A (S_t L_t)^{\phi_j}]^{1/\Delta} - 1.$$

$$g_t = \sum_{i \in \{a, m, s\}} \lambda_{i,t} g_{i,t}^A, \quad 0 < \phi_j < 1.$$

Consumer and producer services share ϕ_c ; $\lambda_{i,t}$ denotes employment shares. Parental education $e_{i,t}$ and cohort skill acquisition p_t^S are distinct choices.

► Skill choice

► Equilibrium

Population and Long-Run Growth

For a terminal skilled share S , fertility converges to replacement:

$$\bar{n}^{\infty} = 1, \quad g^{\infty} = \mathcal{G}(S; \zeta) = \frac{(\gamma\zeta - \tau_s)^2}{\tau_s \tau_e(S)}.$$

Population adjusts to a unique finite level:

$$L^{\infty}(S; \zeta) = \frac{1}{S} \left[\frac{[1 + \mathcal{G}(S; \zeta)]^{\Delta} - 1}{\lambda_A} \right]^{1/\phi_c}.$$

$$\tau_e'(S) < 0 \quad \implies \quad \frac{\partial \mathcal{G}(S; \zeta)}{\partial S} > 0.$$

Terminal service economy; $\tau_h = 0$, $\gamma\zeta > \tau_s$, and fixed full-income ratio $\zeta = 1 + d^{\infty}/\bar{w}^{\infty}$.

► Proof

Manufacturing and Long-Run Growth

Economy A Agriculture $\longrightarrow$ Manufacturing $\longrightarrow$ Services

Economy B Agriculture $\longrightarrow$ Services

$$\begin{aligned} g_A^\infty > g_B^\infty &\iff S_A^\infty > S_B^\infty \iff K_A^\infty > K_B^\infty \\ &\iff \underbrace{\frac{\tau_m}{\tau_s}}_{\text{education}} \underbrace{(S_1 L_1)^{\phi_m - \phi_c}}_{\text{productivity growth}} \underbrace{\left(\frac{F_m}{F_c}\right)^{\frac{2}{1+\phi_E}}}_{\text{firm scale}} > 1. \end{aligned}$$

Common initial state; three-stage comparison with symmetric firms, common prices and terminal full-income ratio, and nonsaturated service formalization; $\tau_h = 0$, $\Delta = 1$.

► Proof and conditions

Calibration

Calibration: The U.S. Development Path

1850–2019; annual model. Joint SMM: 19 parameters, 100 moments in 13 blocks.

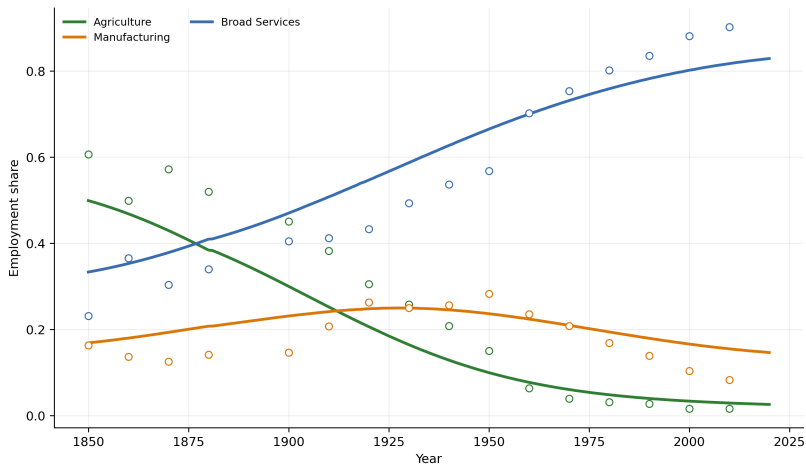
Parameter block	Principal empirical discipline
Demand and productivity	Sectoral employment, manufacturing peak, GDP and VA productivity growth
Fertility and skills	Fertility, college attainment, earnings premium and VA skill shares
Organization: R, ξ_a, ξ_c	Sectoral HSPS employment shares
Entry: F_a, F_m, F_c	Employees per employer firm in agriculture, manufacturing and services

Taking HSPS to the data: IPUMS + BEA Use Table

- Organizational and technical occupations; high skill: bachelor's degree or higher.
- Intermediate-service workers allocated to buyer sectors using BEA expenditure shares; final-consumer sales assigned to consumer services.

Fit and Validation

Fit to U.S. Structural Change

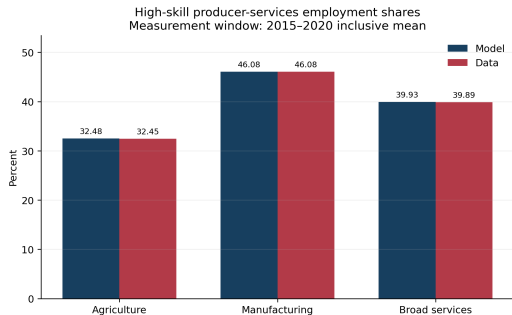


Lines: model; circles: data. Employment paths are targeted. Manufacturing peak: 24.93% in 1928 (model), 28.28% in 1950 (data).

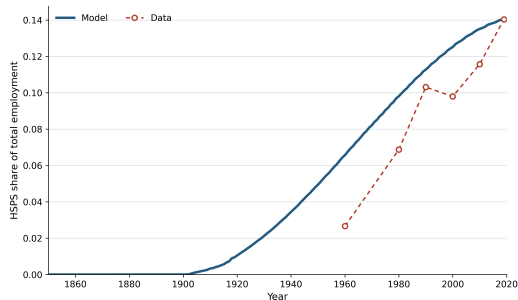
► Targeted fit

HSPS: Modern Fit and Historical Validation

Modern fit



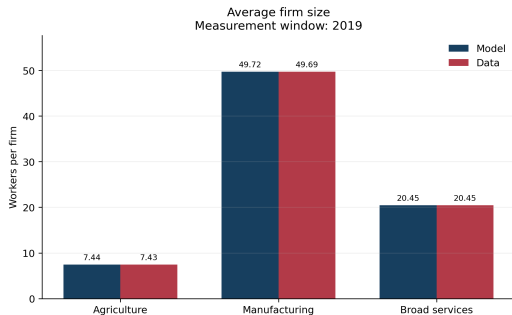
Historical validation



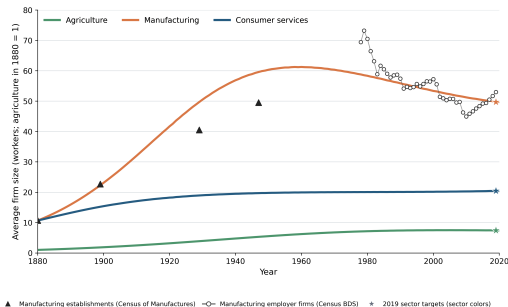
Left: targeted sectoral HSPS shares, 2015–2020 averages. Right: historical HSPS share of total employment. Source: IPUMS and BEA.

Firm Size: Modern Fit and Historical Validation

Modern fit



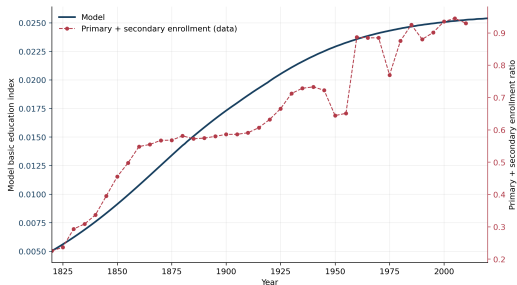
Historical validation



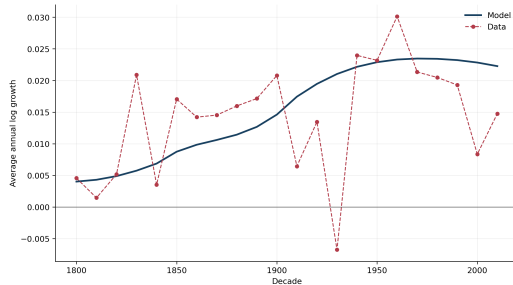
Left: targeted employees per employer firm, 2019. Right: historical manufacturing establishment and employer-firm sizes; initial scale anchored in 1880. Source: U.S. Census Bureau.

Untargeted Validation: Education and Growth

Education



GDP-per-capita growth



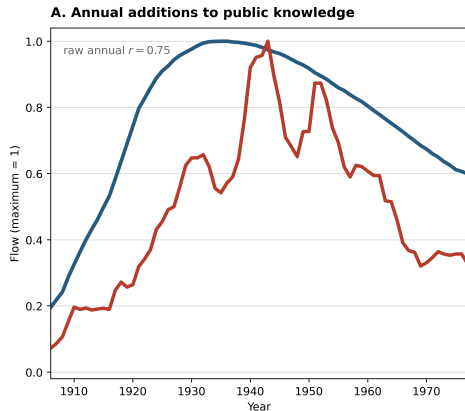
Left: model education index and primary–secondary enrollment. Right: annualized GDP-per-capita growth by decade, with the model shown as a smoothed trend. Neither series is directly targeted; GDP levels enter the calibration.

Measuring Public Organizational Knowledge

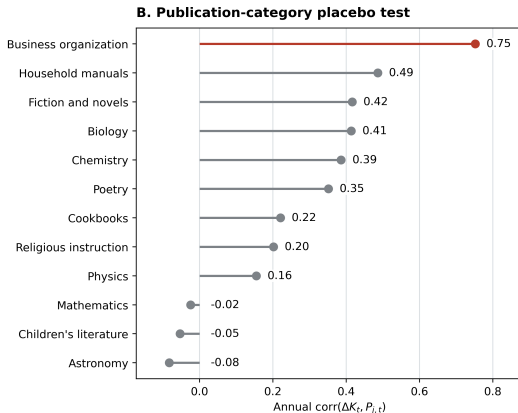
- Extract publications from the Library of Congress *Catalog of Copyright Entries*, 1906–1977.
- Count records containing fixed phrases/words per million OCR words. Compare to ΔK_t for various knowledge categories.

Publication category	Examples of matched words and phrases
Business organization	business systems; office management; personnel management; scientific management; time study; record-keeping...
Household manuals	household management; home economics; domestic science; house-keeping; household hints...
Chemistry	chemistry; chemical science; chemical analysis; biochemistry; electrochemistry...

Untargeted Validation: Public Organizational Knowledge



— Model ΔK — Business-organization publications



Copyright records per OCR word, 1906–1977; explicit manufacturing titles excluded. Lines: normalized five-year averages; correlations: annual data. Source: Library of Congress.

Quantitative Results

Counterfactuals: Manufacturing Histories and Development

1. Vary manufacturing demand

Change only w_m throughout development; common initial conditions and all other parameters fixed.

2. Solve each development path

Manufacturing peaks, organizational knowledge, skills and income emerge in equilibrium.

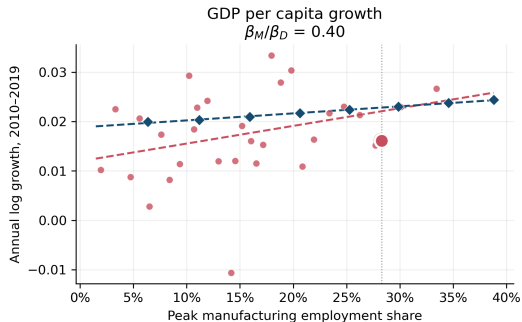
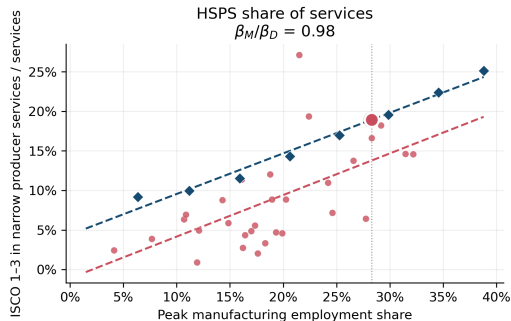
3. Compare model and data

Compare slopes relating each outcome to the historical manufacturing peak.

Outcomes: HSPS, tertiary attainment, income, GDP-per-capita growth and population growth.

► Demand-equivalent wedges

Manufacturing History, HSPS and Service-Era Growth



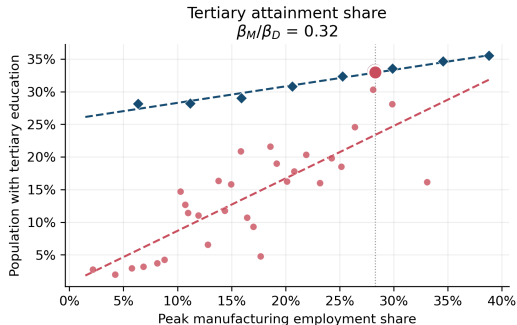
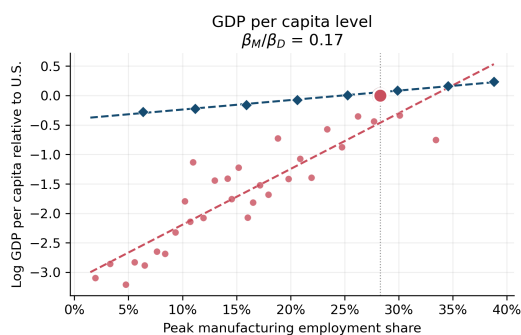
• Data bins ♦ Model economies ● United States

Data: 30 country bins; model: eight economies. Dashed lines: OLS on underlying observations; β_M/β_D : slope ratio.
Samples: 48 countries (HSPS), 172 (GDP).

► HSPS measurement

► Sensitivity

Manufacturing History, Income and Education



• Data bins ♦ Model economies ● United States

Data: 30 country bins; model: eight economies. Income: 2019; tertiary attainment: 2010 and 2015 average. Samples: 172 countries (GDP), 137 (education). Sources: Gapminder and Barro–Lee.

► Sensitivity

Manufacturing History and Subsequent Growth

High peak: 29.88%

Low peak: 11.19%

		2030	2050	2070
Skilled-employment share (%)	High	34.18	34.64	35.02
	Low	28.74	29.37	29.83
GDP-per-capita growth (% p.a.)	High	2.24	2.22	2.20
	Low	1.99	2.01	2.03
GDP per capita (low peak = 100)	High	139.44	145.57	150.89

Only w_m differs: 3.49 (high), 0.60 (low). Growth is the annualized log change in chain-weighted real GDP per capita over the preceding decade.

Mechanism Accounting

Repeat the manufacturing-demand experiment, changing one mechanism at a time.

	Share of baseline slope (%)				
	HSPS share	Tertiary attainment	GDP per capita	GDPpc growth	Population growth
	2010–2019	2010–2019	2019	2010–2019	1850–2019
Baseline	100.0	100.0	100.0	100.0	100.0
Equal entry barriers, F_j	12.7	−38.4	49.0	40.7	36.0
No knowledge diffusion, $\delta_K = 0$	47.2	26.5	71.3	67.6	67.9
Equal productivity elasticities, ϕ_j	119.1	81.8	76.4	118.7	1.5
Equal adaptability, ξ_j	83.6	77.9	92.7	89.7	93.0
Equal VA skill intensity, α_j	99.7	113.0	108.8	99.8	122.8

Common five-economy support; slopes use each counterfactual's realized manufacturing peak. Other parameters fixed; no recalibration.

Industrial Policy

When can a temporary intervention change an economy's growth path?

Recalibrate to a median LMIC with small establishments and scarce skills.

Endogenous establishment setup

Before drawing ability a , entrants choose setup intensity h :

$$x = ha, \quad h \in [\underline{h}, 1].$$

Lower h reduces productivity and the setup inputs needed to enter.

Distortions against productive firms

More productive establishments face larger revenue wedges:

$$1 - \tau_i = \kappa_j x_i^{-d_j}, \quad d_j = 0.41.$$

Removing d_j restores $h = 1$ and the benchmark entry block.

Temporary Support in Different Firm-Scale Environments

Start from the same inherited LMIC state in 2026.

Temporary interventions

Skill acquisition scholarships

Reduce the cost of becoming skilled.

Formalization support

Reimburse the cost of creating a new routine.

Mixed package

Split resources equally between the two.

Comparisons: manufacturing and HSPS wage subsidies.

Permanent firm-scale environments

1. Distortions remain

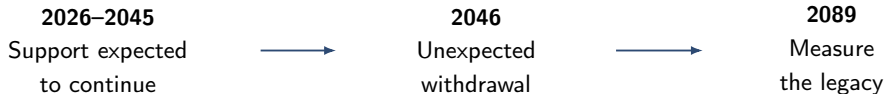
Keep $d_j = 0.41$.

2. Full reform

Reduce d_j to zero over 2026–2035.

3. Reform + U.S. entry costs

Also set F_m, F_c to their U.S. values.



Firm Scale Amplifies the Organizational Legacy

Reform alone, in 2089: firm size roughly doubles; GDP per capita rises 7.6%; public knowledge remains zero.

Additional formalized tasks in 2089 (pp)

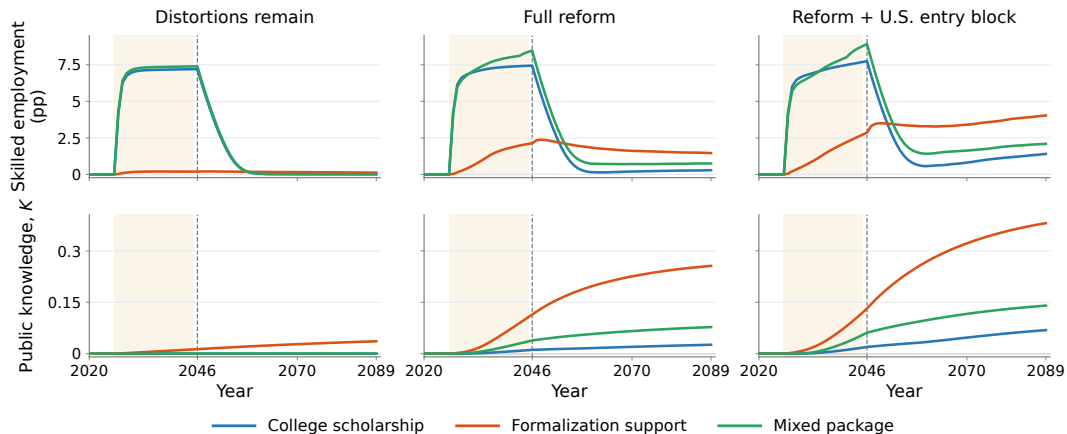
Temporary intervention	Distortions remain	Full reform	Reform + U.S. entry costs
Skill acquisition scholarships	0.00	1.64	9.40
Formalization support	0.27	8.63	26.60
Mixed package	0.00	4.48	13.98
HSPS wage subsidy	0.10	3.15	11.02
Manufacturing wage subsidy	0.00	0.00	0.00

Table: effects relative to each environment's no-support path. Formalization support reaches the reimbursement cap and spends below the fiscal target.

► Costs and welfare

► U.S. entry costs

What Remains after Policy Withdrawal?



Changes relative to each environment's no-support path. Shading: support in 2026–2045; dashed line: unexpected withdrawal in 2046.

Conclusion

1. **Industrial history shapes the service economy.**

Larger manufacturing peaks predict more HSPS and faster subsequent GDP-per-capita growth.

2. **Production at scale builds transferable capabilities.**

Firms develop reusable procedures that diffuse into services and sustain demand for skills.

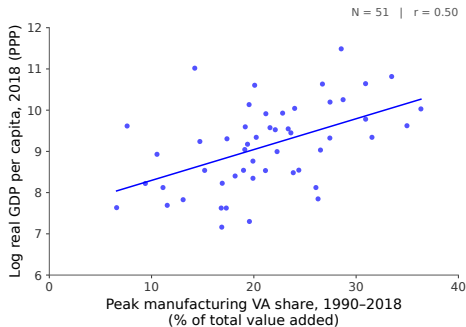
3. **Scale reform and capability policies are complements.**

Temporary support for skill acquisition and formalization has larger persistent effects when firms can grow.

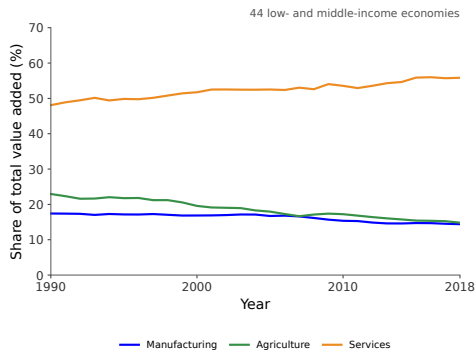
**Development without industrialization requires
other ways to build organizational capabilities.**

Appendix

Motivation: Value Added



(a) Manufacturing VA peaks and income



(b) Sectoral value-added shares

Current-price VA shares, 1990–2018. Left: 51 economies; 2018 PPP income. Right: 44 low/middle-income economies, classified in 2018; unweighted means. Source: ETD and Gapminder.

Manufacturing Peaks and Growth in Service Economies

Dependent variable: average annual GDP-per-capita growth, 2010–2019 (pp).

	All countries		Services > 50%	
	(1)	(2)	(3)	(4)
Manufacturing peak (10 pp)	0.629** (2.53)	0.398 (0.92)	0.645** (1.99)	1.113** (2.36)
Mean service share (10 pp)	−0.572*** (−2.96)	−0.540*** (−2.68)	−1.045** (−2.59)	−1.313** (−2.41)
Mean manufacturing share (10 pp)		0.354 (0.70)		−0.924 (−1.16)
Region and income-group FE	Yes	Yes	Yes	Yes
Institutional quality control	Yes	Yes	Yes	Yes
Observations	173	173	104	104
R^2	0.360	0.362	0.278	0.288

HC1-robust t statistics in parentheses. Income groups: 2009 GNI. Institutions: control-of-corruption percentile. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: GGDC, ILOSTAT and World Bank.

County HSPS Measures and Employment Denominators

Inside manufacturing: occupations

Accounting and record keeping; purchasing, logistics and coordination; personnel and labor relations; technical and scientific work.

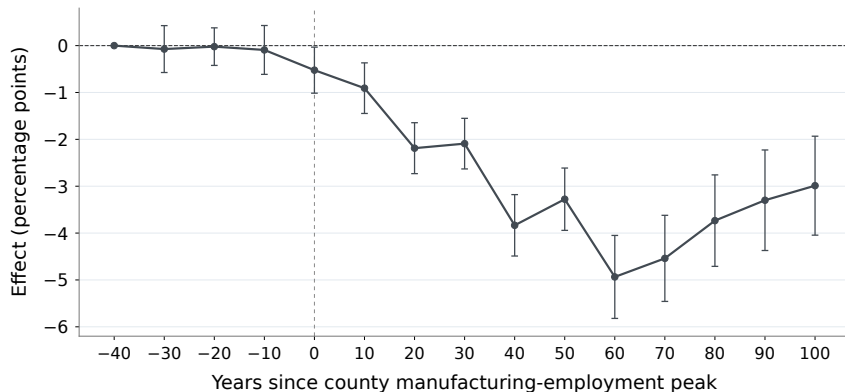
In services: industries

Information (NAICS 51); professional, scientific and technical services (54); management of companies and enterprises (55). Historical Census and SIC data use functional concordances.

Exercise	Denominator
Main event study	Classified employment: primary, manufacturing and services, reconstructed from the numerator's source in each vintage.
2000 cross section	QCEW private-service employment, excluding utilities.

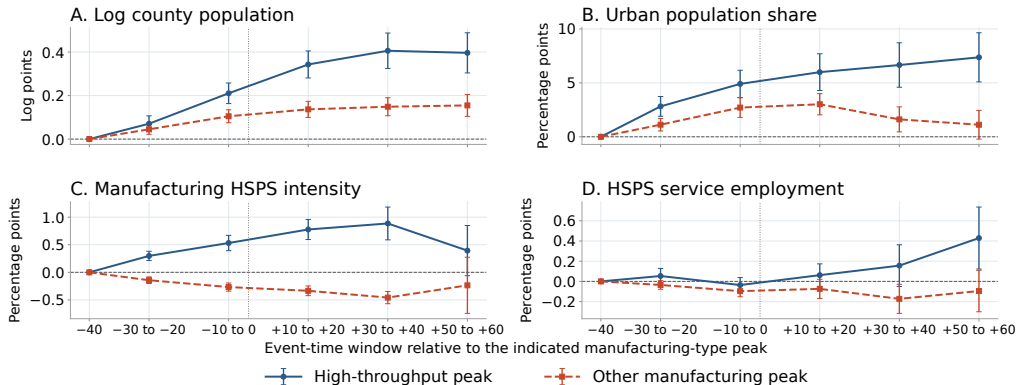
Sources: Census of Manufactures, Population Census, CBP/EFSY and QCEW. Fixed 1850 county geography.

Falsification: Aggregate Service Employment



Covered service employment / classified employment. Same sample and specification as the HSPS-service panel; 95% CIs (Conley, 100 km). Source: Census and CBP.

Manufacturing Type: Development and HSPS Employment



Per 10 pp larger type-specific peak; 95% CIs (Conley, 100 km). HSPS denominators: manufacturing employment (C); classified private employment (D). Source: Census and CBP.

Disclosure Treatment and Labor-Market Geography

High-throughput IV estimates under alternative outcome constructions.

Outcome and geography	<i>N</i>	Estimate	SE	First-stage <i>F</i>	AR <i>p</i>
QCEW public, counties, 2000	1,487	0.283**	0.118	12.87	0.009
QCEW flag-aware, counties, 2001	1,511	0.187*	0.100	11.25	0.066
EFSY native, counties, 2000	1,487	0.103	0.080	12.87	0.202
QCEW flag-aware, CZs, 2001	391	0.324**	0.141	9.03	0.005
EFSY harmonized, CZs, 2001	391	0.204	0.125	9.03	0.094
EFSY harmonized, CZs, 2010	391	0.279**	0.135	9.03	0.009

Preferred controls; Conley SE (100 km). EFSY: imputed CBP employment. CZ: commuting zone. AR: Anderson–Rubin.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

◀ Back

Inference with Common Industry Shocks

Outcome and inference	Estimate	SE	p -value	First-stage F	AR p
<i>High-throughput instrument</i>					
Public QCEW, 2000: county Conley	0.283	0.118	0.016	12.87	0.009
Public QCEW, 2000: exact leave-out	0.283	0.159	0.088	5.26	0.062
Public QCEW, 2000: common shocks	0.284	0.132	0.041	3.94	0.047
Flag-aware QCEW, 2001: exact leave-out	0.187	0.121	0.135	5.02	0.065
Flag-aware QCEW, 2001–03: exact leave-out	0.164	0.108	0.142	5.39	0.082
<i>All-industry instrument</i>					
Public QCEW, 2000: exact leave-out	0.133	0.223	0.551	1.59	0.623

Exact leave-out: industry-clustered scores. Common shocks: leave-sample-out growth. 26 high-throughput industries; effective count: 4.71. Tests use finite-industry degrees of freedom. AR: Anderson–Rubin.

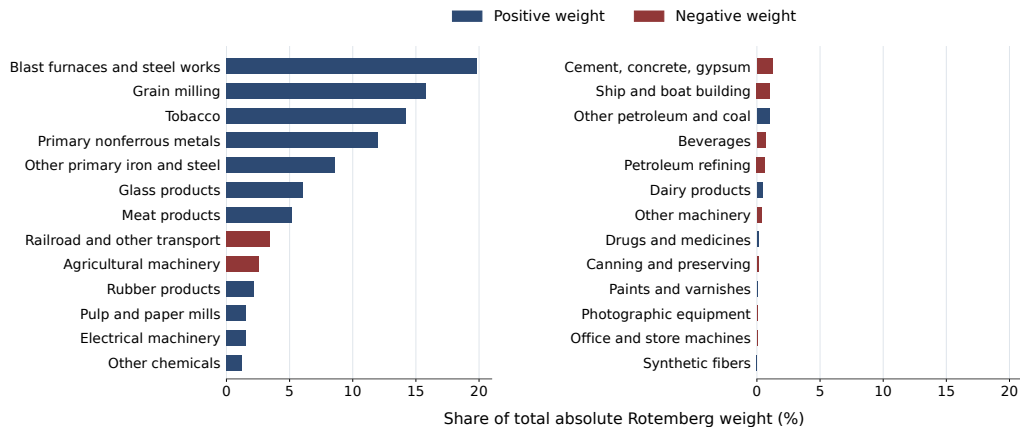
Classification, Exposure, and Conditioning Checks

High-throughput IV; QCEW public-record outcome in 2000.

Specification	Estimate	SE	<i>p</i> -value	First-stage <i>F</i>	AR <i>p</i>
Exclude processed foods	0.325	0.257	0.207	3.63	0.159
Exclude machinery	0.189	0.098	0.055	28.98	0.041
Narrow continuous/process set	0.183	0.078	0.018	29.69	0.011
Add historical geography controls	0.320	0.120	0.008	13.43	0.002
Add manufacturing employment share in 2000	0.196	0.096	0.042	14.81	0.042
Preferred controls; Conley radius 500 km	0.283	0.170	0.096	2.83	0.067

Preferred controls except where indicated. Conley SE, 100 km unless stated otherwise. AR: Anderson–Rubin.

Which Industries Identify the High-Throughput IV?

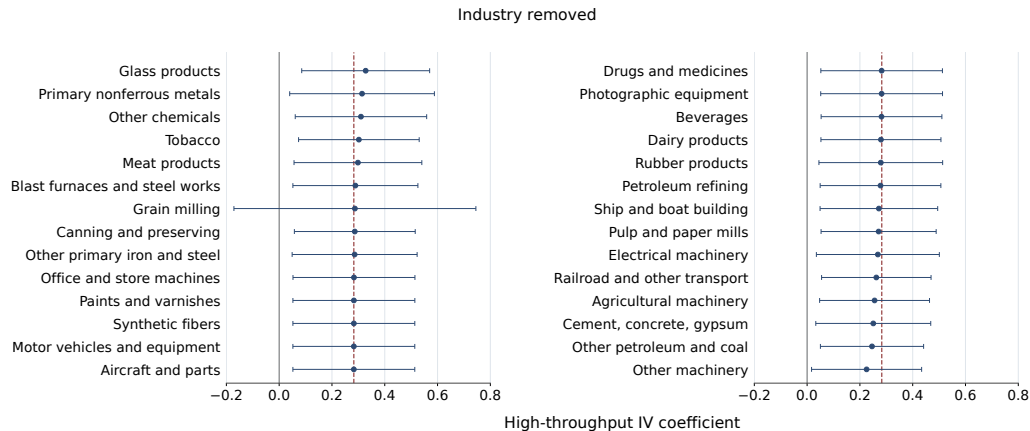


26 industries; absolute Rotemberg weights after residualizing on the preferred controls. The five largest account for 70.4%; negative weights account for 10.2% of total absolute weight. Source: Census and QCEW.

▶ Leave one out

◀ Back

High-Throughput IV: Leave One Industry Out



Each row drops the named industry and rebuilds the shifter. 2SLS estimates with 95% Conley intervals (100 km); preferred controls. The dashed line is the benchmark. Source: Census and QCEW.

◀ Weights

◀ IV results

Manufacturing Peaks and HSPS Employment Growth

Dependent variable: change in HSPS share of total employment, 1990–2018 (pp).

	(1)	(2)	(3)	(4)
Manufacturing peak (10 pp)	2.867*** (0.507)	2.937*** (0.501)	2.173*** (0.782)	2.283** (0.949)
GDP-per-capita gain ($\Delta \ln y$)		−1.202 (0.732)	−0.176 (1.029)	−0.145 (1.079)
Service share, 1990 (pp)				−0.021 (0.068)
Region and initial income-group FE	No	No	Yes	Yes
Economies	51	51	51	51
R^2	0.530	0.550	0.617	0.618

HSPS: business services, finance and real estate. Unweighted OLS; HC3 standard errors in parentheses. Income groups use 1990 GNI. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sources: ETD, GGDC and Maddison (2023).

Production, Producer Services and the Scale Result

$$\ell = \left[\alpha_j (h_s L^s)^{\frac{\sigma_L - 1}{\sigma_L}} + (1 - \alpha_j) (h_u L^u)^{\frac{\sigma_L - 1}{\sigma_L}} \right]^{\frac{\sigma_L}{\sigma_L - 1}}.$$

$$P^P = \left[q^{1/\sigma_P} (H^P)^{\frac{\sigma_P - 1}{\sigma_P}} + (1 - q)^{1/\sigma_P} (L^P)^{\frac{\sigma_P - 1}{\sigma_P}} \right]^{\frac{\sigma_P}{\sigma_P - 1}}, \quad \omega^P(q) = \left[q(p^H)^{1 - \sigma_P} + (1 - q)(p^L)^{1 - \sigma_P} \right]^{\frac{1}{1 - \sigma_P}}.$$

$$\ell^*(a, q) = \left[\frac{\eta p^\nu(q) A a}{w^\nu + c^o(q)} \right]^{\frac{1}{1 - \eta}}, \quad n^*(a, q) = \nu \ell^*(a, q).$$

The envelope theorem and the labor first-order condition imply

$$\frac{\partial \pi^*}{\partial q} = \nu^* \frac{\partial p^\nu}{\partial q} + \frac{n^*}{\bar{N}} (r p^L - p^H) = n^* \left[\frac{w^\nu + c^o(q)}{\eta \nu} \frac{\partial \log p^\nu(q)}{\partial q} + \frac{r p^L - p^H}{\bar{N}} \right].$$

Sector and time subscripts suppressed. Competitive producer-service supply gives $p^H = w^s/A^H$ and $p^L = w^u/A^L$.

Organizational Investment, Diffusion and Entry

$$C_{jt}^O(s, u; K_t) = \int_s^u \tilde{C}_{jt}(z; K_t) d\mathcal{F}(z), \quad \tilde{C}_{jt}(z; K_t) = \begin{cases} \min\{C_t^C(z), C_{jt}^A(z)\}, & \mathcal{F}(z) \leq K_t, \\ C_t^C(z), & \mathcal{F}(z) > K_t. \end{cases}$$

$$\mathcal{E}_t = \frac{1}{N_t} \sum_j \int n_{jt}^*(a, s) \left[\int_{\mathcal{F}(z) > K_t} c_{jt}(a, s; z) d\mathcal{F}(z) \right] d\mu_{jt}(a, s).$$

$c_{jt}(a, s; z)$ indicates a newly created proprietary procedure.

$$N_t = \sum_j \int n_{jt}^*(a, s) d\mu_{jt}(a, s), \quad K_{t+1} = \min\{1, K_t + \delta_K \mathcal{E}_t\}.$$

$$I_{jt} \geq 0, \quad F_j I_{jt}^{\phi_E} - V_{jt}^E \geq 0, \quad I_{jt} [F_j I_{jt}^{\phi_E} - V_{jt}^E] = 0.$$

Household Decision Rules and Skill Acquisition

Fertility and continuous education

$$e_{i,t}^* = \left[\sqrt{g_t \left(\frac{\tau_i}{\tau_e(S_t)} - \tau_h \right)} - \tau_h \right]_+, \quad n_{i,t} = \frac{\gamma \zeta_t}{\tau_i + \tau_e(S_t) e_{i,t}^*}, \quad \zeta_t = 1 + \frac{d_t}{\bar{w}_t}.$$
$$\bar{n}_t = \sum_i \lambda_{i,t} n_{i,t}, \quad \tau_e(S_t) = 1 - S_t(1 - \tau_e^s).$$

Discrete skill choice of entering cohorts

$$p_t^s = \left[1 + \exp \left(- \frac{\beta_\pi \log \Pi_{t+1} - c^{\text{skill}}}{\sigma_{\text{skill}}} \right) \right]^{-1}.$$

c^{skill} : skill-acquisition cost; σ_{skill} : dispersion of idiosyncratic choice costs.

$[x]_+ = \max\{x, 0\}$. Education rule shown under $\tau_i/\tau_e(S_t) > \tau_h$; firms' organizational education input uses education chosen 30 years earlier.

Perfect-Foresight Competitive Equilibrium

Given initial population, skills, knowledge, productivities and firm distributions:

1. Households choose consumption, fertility, education and cohort skill.
2. Firms choose inputs, employment and next-period formalization.
3. Entry satisfies free entry; producer-service suppliers are competitive.
4. Final goods, both producer services and both labor markets clear.
5. Aggregate states and firm distributions follow their laws of motion.
6. Anticipated aggregate prices and states equal their realizations.

$$\mu_{j,t+1}(B) = (1 - \delta_f) \int 1\{(a, u_{jt}^*(a, s)) \in B\} d\mu_{jt}(a, s) + I_{j,t+1} \int 1\{(a, 0) \in B\} dG(a).$$

Manufacturing is the numeraire. Idiosyncratic ability and exit risk; no aggregate uncertainty. Transition solved by backward firm optimization and forward aggregation.

Proposition 2: Replacement Fertility and Growth

Fix S and ζ in the terminal service economy; set $\tau_h = 0$.

$$g(L) = [1 + \lambda_A(SL)^{\phi_c}]^{1/\Delta} - 1, \quad e(L) = \sqrt{\frac{\tau_s g(L)}{\tau_e(S)}}, \quad n(L) = \frac{\gamma \zeta}{\tau_s + \tau_e(S)e(L)}.$$

$g(L)$ is increasing and $n(L)$ decreasing, with $n(0) > 1$ and $n(\infty) = 0$.

$$T(L) = Ln(L)^{1/\Delta}, \quad 0 < 1 - \frac{\phi_c}{2\Delta} < \frac{d \log T(L)}{d \log L} < 1.$$

At the unique fixed point, $n = 1$:

$$e^\infty = \frac{\gamma \zeta - \tau_s}{\tau_e(S)}, \quad g^\infty = \frac{(\gamma \zeta - \tau_s)^2}{\tau_s \tau_e(S)}, \quad \frac{\partial g^\infty}{\partial S} > 0.$$

$\gamma \zeta > \tau_s$, $0 < \phi_c < 1$ and $\Delta \geq 1$. The population map is increasing and moves toward the fixed point from either side.

Proposition 3: The Manufacturing Legacy

Under the three-stage comparison:

$$K_{A,3} > K_{B,3} \iff \bar{e}_m^o n_m > \bar{e}_c^o n_c.$$

$$\frac{\bar{e}_m^o}{\bar{e}_c^o} = \left[\frac{\tau_m}{\tau_s} (S_1 L_1)^{\phi_m - \phi_c} \right]^{1/2}, \quad \frac{n_m}{n_c} = \left(\frac{F_m}{F_c} \right)^{\frac{1}{1+\phi_E}}.$$

In the terminal service economy, additional public procedures raise formalization on an unsaturated margin:

$$K_{A,3} > K_{B,3} \iff S_A^\infty > S_B^\infty \iff g_A^\infty > g_B^\infty.$$

Common initial states, prices, operating-surplus terms and terminal full-income ratio; symmetric static firms; $\tau_h = 0$, $\Delta = 1$. Nonsaturation requires that an additional public procedure changes formalization for some service firms.

HSPS: Measurement in Calibration and Cross-Country Data

U.S. calibration: use-based sectoral allocation

- In-house organizational and technical occupations, plus purchased producer services.
- Bachelor's degree or higher defines high skill.
- BEA intermediate-use expenditure shares allocate service suppliers to destination sectors; final sales enter consumer-service value added.

Cross-country accounting: harmonized industry–occupation proxy

ISCO-08 occupations 1–3 in ISIC divisions 62–66, 69–75
Employment in market services: ISIC sections G–N, R–S

ILOSTAT, 2010–2019 averages; 48 countries.

The U.S. allocation assumes common skill composition and labor requirements across an industry's buyers. Aggregate HSPS is an accounting aggregate of sectoral targets.

◀ Calibration

◀ Cross-country results

Calibration: Parameter Estimates and External Benchmarks

Block	Jointly estimated parameters	Estimates
Fertility	τ, τ_a^n	0.0324, 0.4402
Productivity	$\phi_a, \phi_m, \phi_c, g_{GO}$	0.9984, 0.9192, 0.6905, 0.00318
Skills	$c_{col}, \alpha_a, \alpha_m, \alpha_c$	0.5682, 0.2434, 0.3300, 0.4100
Demand	ϵ_c, w_a, w_c	1.0105, 0.6510, 3.8233
Organization	R, ξ_a, ξ_c	22.79, 3.5640, 6.8774
Entry	F_a, F_m, F_c	25.872, 187.625, 85.117

Selected assigned parameters	Value / empirical anchor
δ_K	0.09; diffusion benchmark (Mansfield, 1961)
ρ	0.40; maintained adaptation/creation cost ratio
$\bar{N}$	25; recurring coordination uses about 5% of employment
σ_{skill}	0.10; college attainment and earnings-premium changes
$\eta; \sigma$	0.85 (span of control); 0.30 (final-good substitution)

Joint SMM. The nonredundant Jacobian has rank 19; effective rank 17 at a 1% cutoff. Service adaptability is weakly pinned down within its saturation region.

Calibration: Targeted Scalar Moments

Moment	Data	Model
HSPS share: agriculture	0.3245	0.3248
HSPS share: manufacturing	0.4608	0.4608
HSPS share: consumer services	0.3989	0.3993
Employees per firm: agriculture	7.435	7.441
Employees per firm: manufacturing	49.694	49.717
Employees per firm: services	20.454	20.454
VA productivity growth: agriculture	0.0371	0.0385
VA productivity growth: manufacturing	0.0317	0.0298
VA productivity growth: consumer services	0.0142	0.0131
Aggregate skilled share	0.3299	0.3247
Skilled earnings premium	1.600	1.645
Peak manufacturing-employment share	0.2828	0.2493
Agriculture/service fertility ratio	2.000	1.990

Shares and annual growth rates in fractions. HSPS: 2015–2020; firm size: 2019; productivity growth: 1970–2010. Employment, fertility and GDP paths also enter SMM.

◀ Calibration

◀ U.S. fit

Manufacturing Demand and Equivalent Consumption Wedges

At fixed income and producer prices, define

$$f_m \equiv \frac{w_m^{\text{CF}}}{w_m^*}, \quad k_j \equiv \frac{\varepsilon_j}{\varepsilon_m}, \quad j \in \{a, c\}.$$

Match the relative-demand terms

$$\underbrace{\left(\frac{p_j}{p_m}\right)^{1-\sigma} \frac{w_j}{(f_m w_m^*)^{k_j}}}_{\text{Demand shifter}} = \underbrace{\left(\frac{(1+\tau_j)p_j}{p_m}\right)^{1-\sigma} \frac{w_j}{(w_m^*)^{k_j}}}_{\text{Consumption wedge}}$$

$$(1+\tau_j)^{1-\sigma} = f_m^{-k_j} \iff \boxed{1+\tau_j = f_m^{-k_j/(1-\sigma)}}.$$

◀ Experiment

◀ Growth paths

Sensitivity of the Cross-Country Gradients

18 economic variants: all 90 slopes retain their benchmark sign.

	GDP level	GDP growth	Tertiary	HSPS	Population growth
Median absolute change (%)	2.9	8.5	10.2	2.4	4.3
<i>Exact diffusion grid: $\delta_K \in [0.04, 0.14]$</i>					
Minimum change (%)	-13.8	-6.9	-27.8	-22.4	-16.2
Maximum change (%)	0.6	5.0	4.1	1.6	0.6

- Broad variants: adaptation costs, diffusion, scale, substitution, ability dispersion, education costs and knowledge accumulation.
- All signs also survive the wider diffusion grid $\delta_K \in [0.01, 0.15]$.

No recalibration. Broad variants use a common coarser grid. At $\delta_K = 0.02\text{--}0.05$, one economy per row has residual discrete-formalization oscillation; precise magnitudes are qualified.

◀ HSPS and growth

◀ Mechanism accounting

Temporary Policies: Costs and Outcomes

Temporary policy	PV cost (% GDP)	CEV (%)	GDPpc (%)	Formalized tasks (pp)	Skilled emp. (pp)	ΔK (level)
Distortions remain						
Skill acquisition scholarships	0.593	2.59	5.56	0.00	0.01	0.000
Formalization support	0.017*	0.18	2.13	0.27	0.13	0.036
Mixed package	0.613	6.95	8.02	0.00	0.02	0.000
HSPS wage subsidy	0.101 [†]	−3.31	−1.51	0.10	0.04	0.009
Manufacturing wage subsidy	0.598	−3.48	−2.14	0.00	−0.01	0.000
Full permanent distortion reform						
Skill acquisition scholarships	0.581	6.92	8.52	1.64	0.29	0.026
Formalization support	0.123*	1.67	8.43	8.63	1.46	0.256
Mixed package	0.600	7.51	10.78	4.48	0.75	0.078
HSPS wage subsidy	0.580	0.57	1.79	3.15	0.55	0.052
Manufacturing wage subsidy	0.597	−0.16	−0.42	0.00	0.00	0.000

PV cost: discounted spending / discounted GDP in 2026–2045. CEV: 2026–2089; other outcomes: 2089, relative to each environment's no-policy path. *100% reimbursement cap. [†]Fiscal spending jumps from 0.101% to 2.083% across the threshold.

Full Reform with the U.S. Entry Block

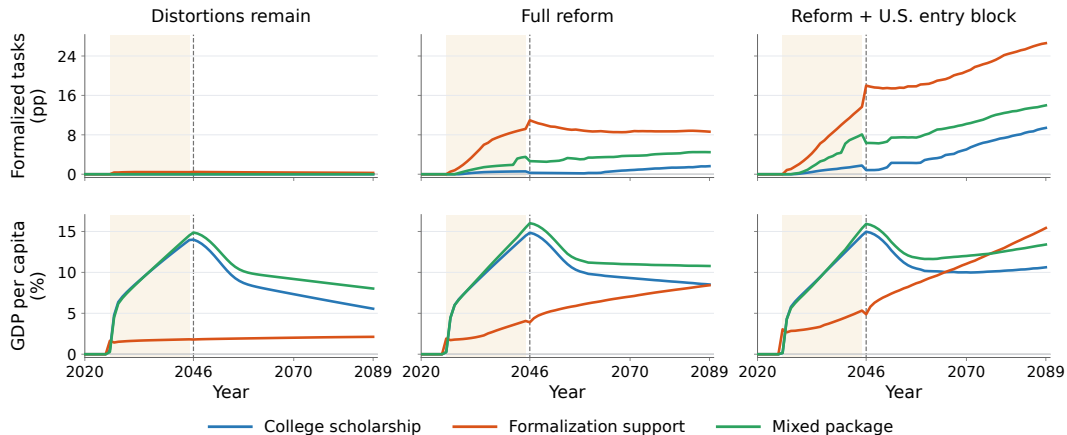
From the common inherited 2026 state: $F_m = 187.625$, $F_c = 85.117$.

Temporary policy	PV cost (% GDP)	CEV (%)	GDPpc (%)	Formalized tasks (pp)	Skilled emp. (pp)	ΔK (level)
Skill acquisition scholarships	0.579	7.17	10.62	9.40	1.41	0.069
Formalization support	0.130*	2.30	15.43	26.60	4.04	0.381
Mixed package	0.585	7.77	13.40	13.98	2.10	0.140
HSPS wage subsidy	0.620	0.70	3.82	11.02	1.63	0.086
Manufacturing wage subsidy	0.597	-0.15	-0.41	0.00	0.00	0.000

Policies operate in 2026–2045. CEV: 2026–2089; other outcomes: 2089. Effects are relative to the no-policy path with full reform and U.S. entry costs. * Formalization support reaches the 100% reimbursement cap.

◀ Policy results

Policy Withdrawal: Formalization and Income



Changes relative to each environment's no-policy path, 2020–2089. Shading: temporary support in 2026–2045; dashed line: unexpected withdrawal in 2046.